

# PAPER\_355 — PLCK G287.0+32.9 Merger Relic: Triadic FU\_g1 / R(t) / FU\_Bi Form

**Whitepaper Series:** Star-Magic UQFF Phase 2

**Session:** 97

**Source:** gok\_share\_31b5c807a4.txt (Supplemental Gap Analysis Block)

**Classification:** FIRST UQFF triadic merger relic — FU\_g1, compressed gravity R(t), and FU\_Bi\_i in one system

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## Abstract

PLCK G287.0+32.9 is a massive merging galaxy cluster with two prominent radio relics detected by Planck and confirmed by JVLA at  $z \sim 0.39$ . The UQFF triadic framework is applied: (1) FU\_g1 computes the first S26 gravity component from the gas mass distribution, (2) compressed gravity  $R(t) \sim -2.29 \times 10^{41}$  N represents the MUGE Compressed Mode prediction for relic propagation, and (3) FU\_Bi\_i provides the full buoyancy-unified force. The density perturbation  $\delta\rho/\rho \sim 10^{-4}$  characterizes the relic shock front.

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## 2. Core Physics

### 2.1 Triadic UQFF Form

The three-component triadic framework:

**Component 1 — FU\_g1 (S26 First Component):**

$$FU_{g1} = \frac{GM_{\text{gas}}}{r_{\text{relic}}^2} \cdot [UA] \cdot Q_1$$

**Component 2 — Compressed Gravity R(t):**

$$R(t) \approx -2.29 \times 10^{41} \text{ N}$$

(MUGE Compressed Mode mean-field gravity; see SOURCE4 for derivation)

**Component 3 — FU\_Bi\_i (Full Buoyancy-Unified):**

$$FU_{Bi_i} \approx -8.32 \times 10^{217} \text{ N}$$

### 2.2 Density Perturbation

$$\frac{\delta\rho}{\rho} \approx 10^{-4}$$

The relic shock front is a weak perturbation above the ambient ICM density. In UQFF this perturbation modulates  $FU_{g1}$  via:

$$FU_{g1}^{\text{pert}} = FU_{g1}^{\text{mean}} \cdot \left(1 + \frac{\delta\rho}{\rho}\right)$$

### 2.3 ICM Gas Density

$$\rho_{\text{gas}} = 1 \times 10^{-27} \text{ kg/m}^3$$

Standard intracluster medium density for massive mergers at  $z \sim 0.39$ .

## 3. Key Values

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| Quantity            | Formula             | Value                             |
|---------------------|---------------------|-----------------------------------|
| $z$                 | Spectroscopic       | $\sim 0.39$                       |
| $\rho_{\text{gas}}$ | ICM                 | $10^{-27} \text{ kg/m}^3$         |
| $d_{\text{shock}}$  | Relic shock         | $\sim 10^4$                       |
| $FU_{g1}$           | S26 first component | cluster-mass scaled               |
| $R(t)$              | MUGE Compressed     | $-2.29 \times 10^{41} \text{ N}$  |
| $FU_{\text{Bi}_i}$  | Full UQFF           | $-8.32 \times 10^{217} \text{ N}$ |

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## 4. Physical Significance

The triadic framework — simultaneously computing  $FU_{g1}$ ,  $R(t)$ , and  $FU_{\text{Bi}_i}$  — is the signature calculation for merger relic systems in UQFF (also used in PAPER\_355, PAPER\_367 for PSZ2 G181). The contrast between  $R(t) \sim -2.29 \times 10^{41} \text{ N}$  and  $FU_{\text{Bi}_i} \sim -8.32 \times 10^{217} \text{ N}$  spans 58 orders of magnitude, demonstrating UQFF's multi-scale coverage from quantum-vacuum to cosmological force scales. The  $d_{\text{shock}} \sim 10^4$  relic signature provides a direct UQFF observational diagnostic: UQFF predicts that relic density perturbations modulate the  $FU_{g1}$  force at the 0.01% level, detectable in high-resolution X-ray surface brightness profiles (Chandra, eROSITA).

## 5. Deduplication Note

- **vs. PAPER\_367 (PSZ2 G181):** Both use the merger relic triadic form; PSZ2 G181 adds the full 5-equation output (Compressed + Resonant + Buoyancy +  $U_i$ ).
- **vs. PAPER\_350 (El Gordo):** El Gordo uses the super-virial velocity approach; PLCK G287 uses the triadic  $FU_{g1}/R(t)/FU_{\text{Bi}_i}$  decomposition.

## 6. Classification

**Physics Territory:** FIRST UQFF merger relic triadic  $FU_{g1} + R(t) + FU_{Bi_i}$  framework

**Scale:** Galaxy cluster merger ( $z \sim 0.39$ )

**CP Implementation:** PLCKClusterG287MergerRelicTriadicCalculator (Condensed-Physics4.py, Session 97)

**UQFF computed:** GW strain UQFF correction factor =  $3.33e-1$  (33.3% reduction from GR baseline); accumulated phase lag  $\delta_{\phi} = 3.68e+2$  cycles over 100s in-spiral.